



Analyzing Data Packets Using Wireshark

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

Wireshark is a widely used network protocol analyzer that captures and displays detailed information about network packets. It is commonly used to study protocols, troubleshoot network issues, and detect threats such as attacks and Trojan horses. After completing this task, you will gain a foundational understanding of Wireshark and be able to use Wireshark to analyze network packets.

1.2 Objectives

On completing this task, you will be able to:

1. Be familiar with and master basic Wireshark operations, and understand the interaction and packet exchange between network protocol entities.
2. Know how to use capture filters and display filters in Wireshark.
3. Analyze Wireshark packet capture instances to deepen understanding of network protocols.

2 Task Preparation

2.1 Network Topology

The network topology of this task requires only one host that has Internet access.

Figure 1 Lab topology



2.2 Initial Configuration

Before using Wireshark to capture packets, you need to download and install Wireshark. The installation is straightforward. Simply follow the prompts and click **Next** to complete the process. The Wireshark official website is <https://www.wireshark.org/>.



3 Task Implementation

3.1 Capturing Packets Using Wireshark

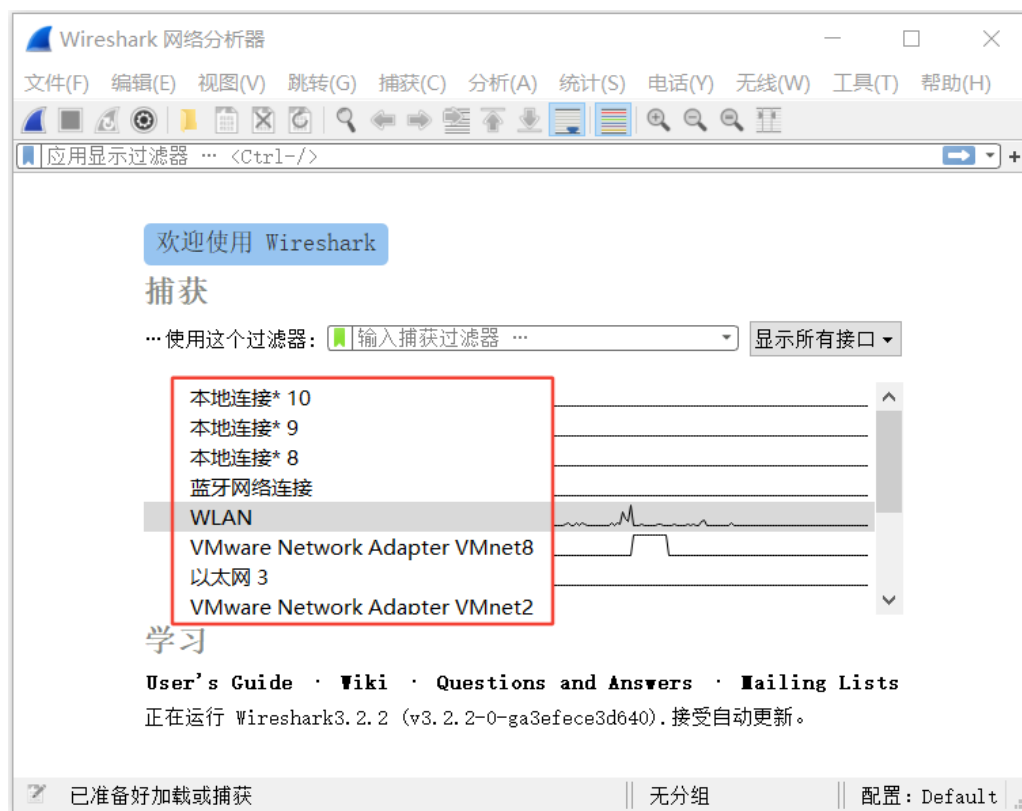
Step 1 Start Wireshark, as shown in Figure 2.

Figure 2 Starting Wireshark



Step 2 After Wireshark is started, all network adapters available in the system are listed in the main window of Wireshark, as shown in Figure 3.

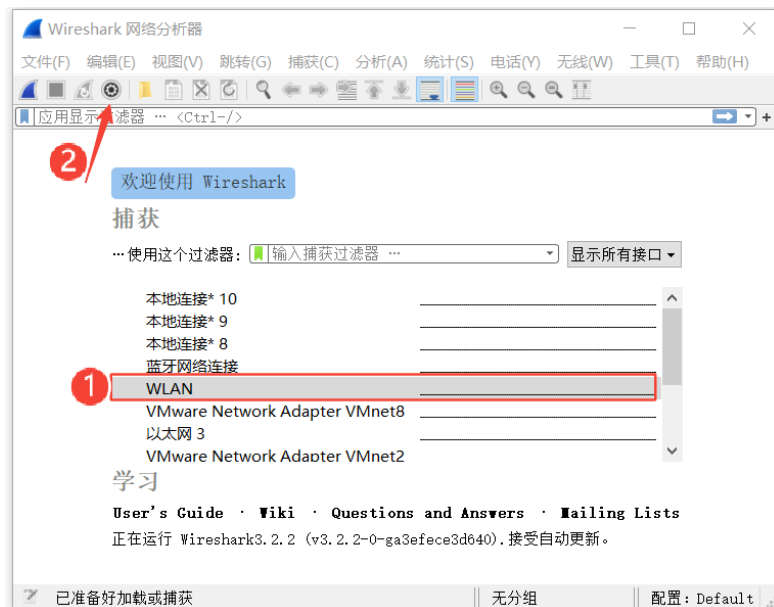
Figure 3 Network adapter information





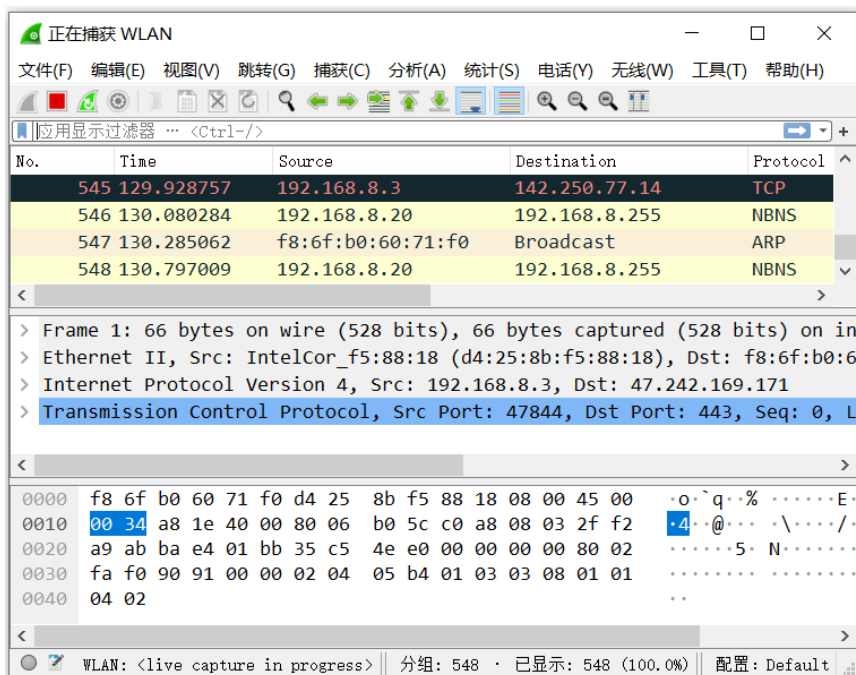
Select the network adapter to be listened on (**WLAN** is selected in this example, as shown in Figure 3), and double-click it to enter the listening mode. You can also click the button **Options...** in the main toolbar, as shown in Figure 4.

Figure 4 Clicking the button to enter the listening mode



Step 3 Double-click the selected network adapter to start packet capture. Wireshark then starts to capture packets, as shown in Figure 5.

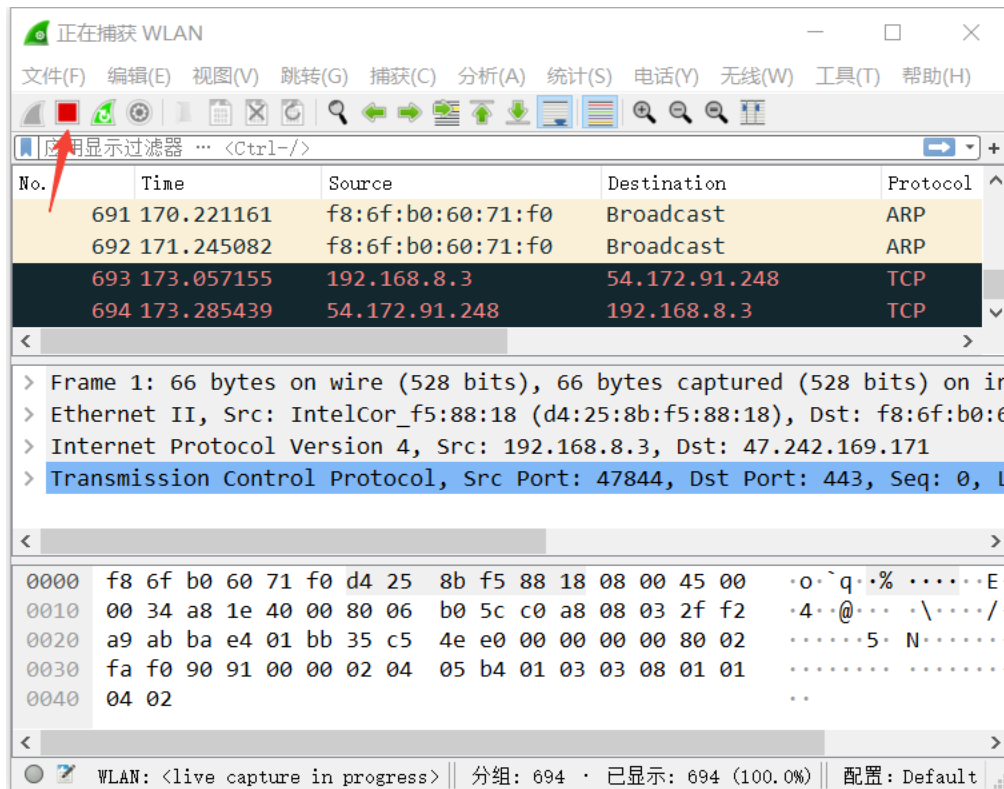
Figure 5 Starting packet capture





During the packet capture process, you can find that the packet information changes. To stop packet capture, click the button **Stop** as shown in Figure 6.

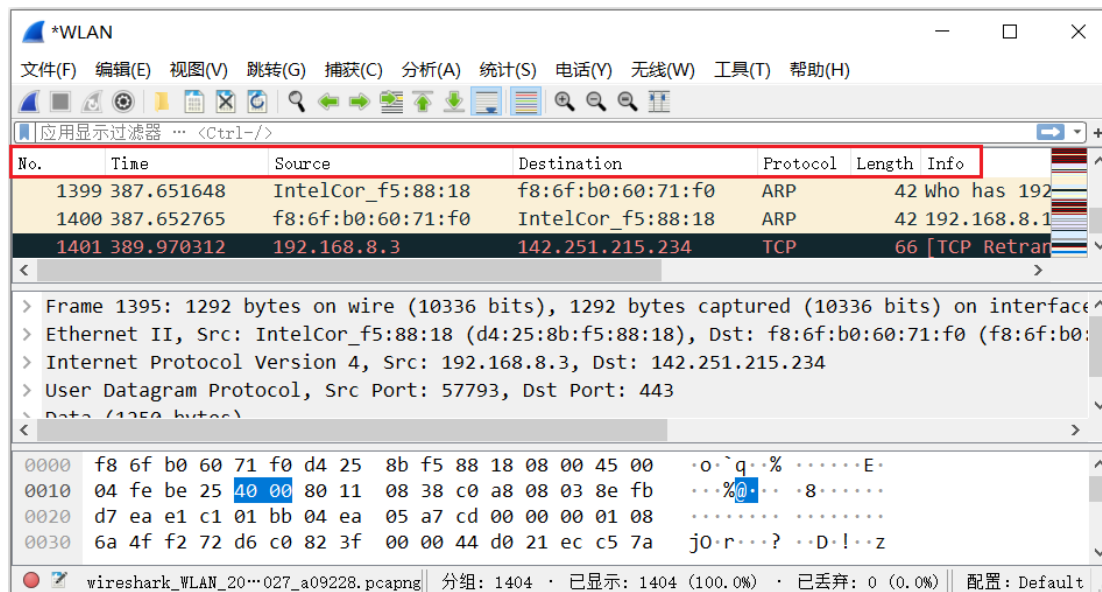
Figure 6 Stopping packet capture



The main window of Wireshark displays the packet list, which includes the sequence number, time, source IP address, destination IP address, protocol, length, and basic information. Wireshark uses different colors to distinguish protocols, as shown in Figure 7.



Figure 7 Main window of Wireshark



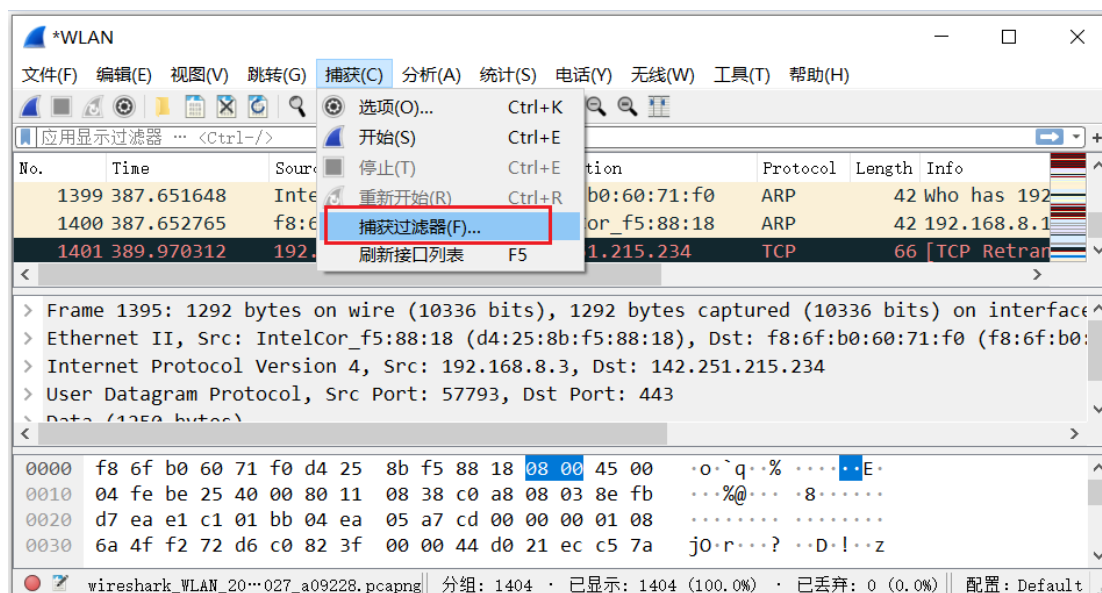
In most cases, a large number of packets are captured. Wireshark provides two types of filters: capture filters and display filters.

3.2 Configuring a Capture Filter

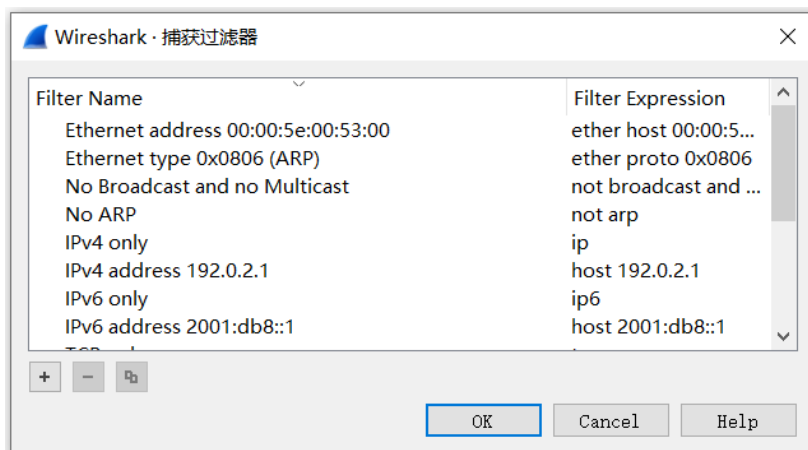
A capture filter is used to specify the type of packets to be captured. It must be set before starting the capture process.

Step 1 Choose **Capture > Capture Filters**, as shown in Figure 8.

Figure 8 Selecting a capture filter



The capture filter page is displayed, as shown in Figure 9.

**Figure 9** Capture filter page

On the capture filter page, you can view, modify, add, and delete filters as needed.

Step 2 Understand the syntax of capture filters. Figure 10 shows the syntax.

Figure 10 Syntax of capture filters

Syntax	Protocol	Direction	Host(s)	Value	Logical Operation	Other Expression
Example	tcp	dst	1.1.1.1	80	and	tcp dst 2.2.2.2 1560

- **Protocol:** The possible values include **ether**, **fdi**, **ip**, **arp**, **rarp**, **decnet**, **lat**, **sca**, **moprc**, **mopdl**, **tcp**, and **udp**. If it is not specified, all supported protocols are included.
- **Direction:** The possible values include **src**, **dst**, **src and dst**, and **src or dst**. The value **src or dst** is used by default if no direction is specified.
- **Host(s):** The possible values include **net**, **port**, **host**, and **portrange**. The value **host** is used by default if it is not explicitly defined.
- **Logical Operations:** The possible values include **not**, **and**, and **or**. The operator **not** has the highest priority. When the operators **and** and **or** appear together, they have the same priority and are evaluated from left to right.

Examples:

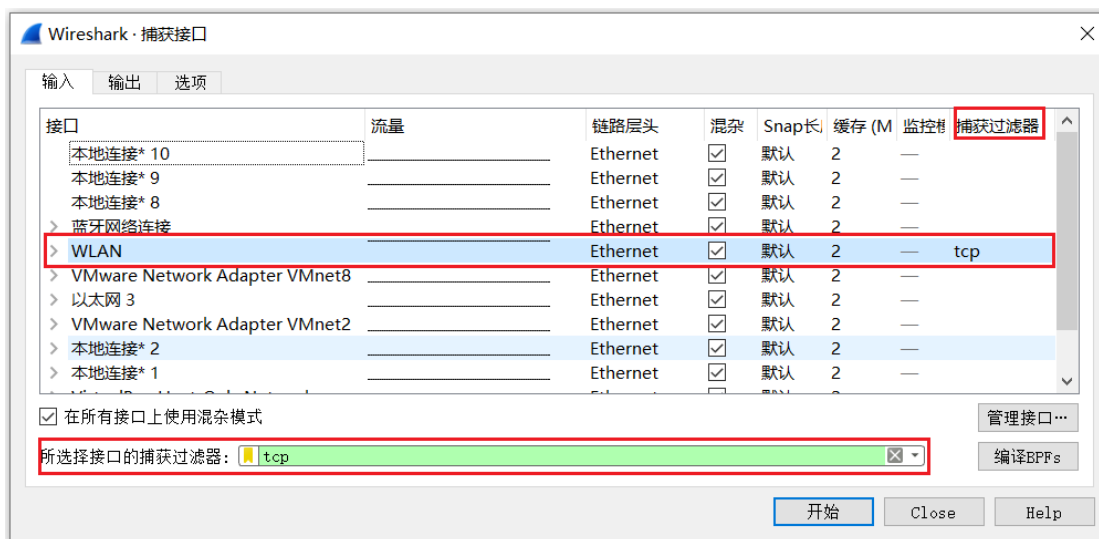
- **tcp dst port 1560:** captures packets whose destination port is TCP port 3128.
- **ip src host 10.1.1.1:** captures packets whose source IP address is 10.1.1.1.
- **host 10.1.2.3:** captures packets whose destination or source IP address is 10.1.2.3.
- **src portrange 2000-2500:** captures packets whose source port is a UDP or TCP port in the range from 2000 to 2500.
- **not icmp:** captures all packets except ICMP packets. (ICMP is usually used for ping operations.)
- **src host 10.7.2.12 and not dst net 10.200.0.0/16:** captures packets whose source IP address is 10.7.2.12 but the destination is not 10.200.0.0/16.

When using a keyword as a value, prefix it with a backslash (\). **ether proto lip**, equivalent to the keyword **ip**, captures IP packets. **ip proto licmp**, equivalent to the keyword **icmp**, captures IP packets of the ICMP protocol. You can also append the keywords **multicast** and **broadcast** after **ip** or **ether**. To exclude broadcast requests, use **no broadcast**.



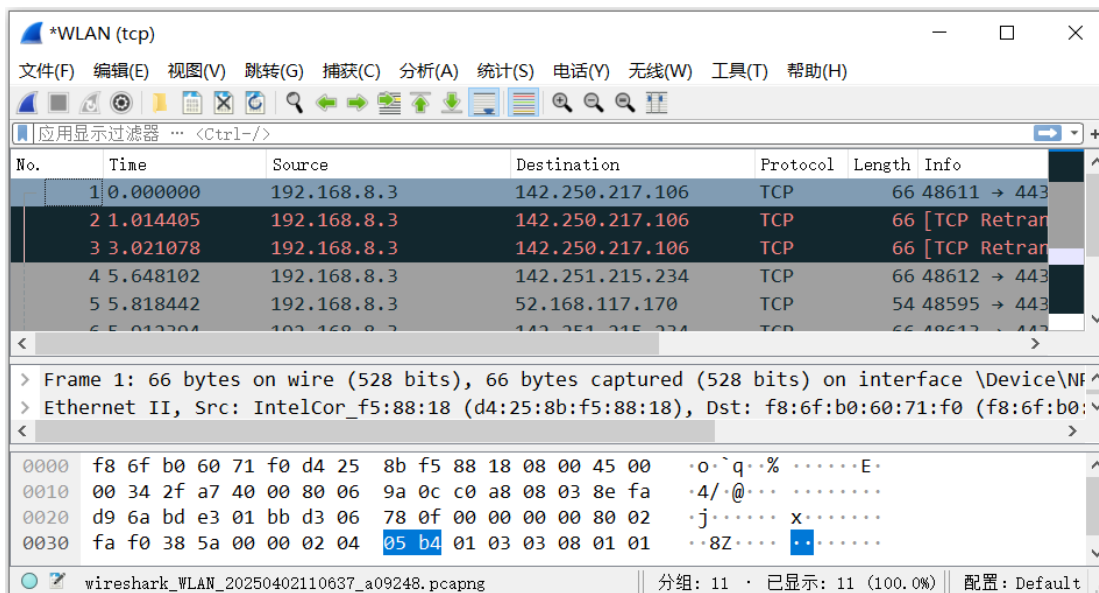
Step 3 Configure a capture filter to capture only TCP packets, as shown in Figure 11.

Figure 11 Configuring a capture filter to capture TCP packets



Click **Start**. Wireshark starts to capture TCP packets only, as shown in Figure 12.

Figure 12 Capturing TCP packets



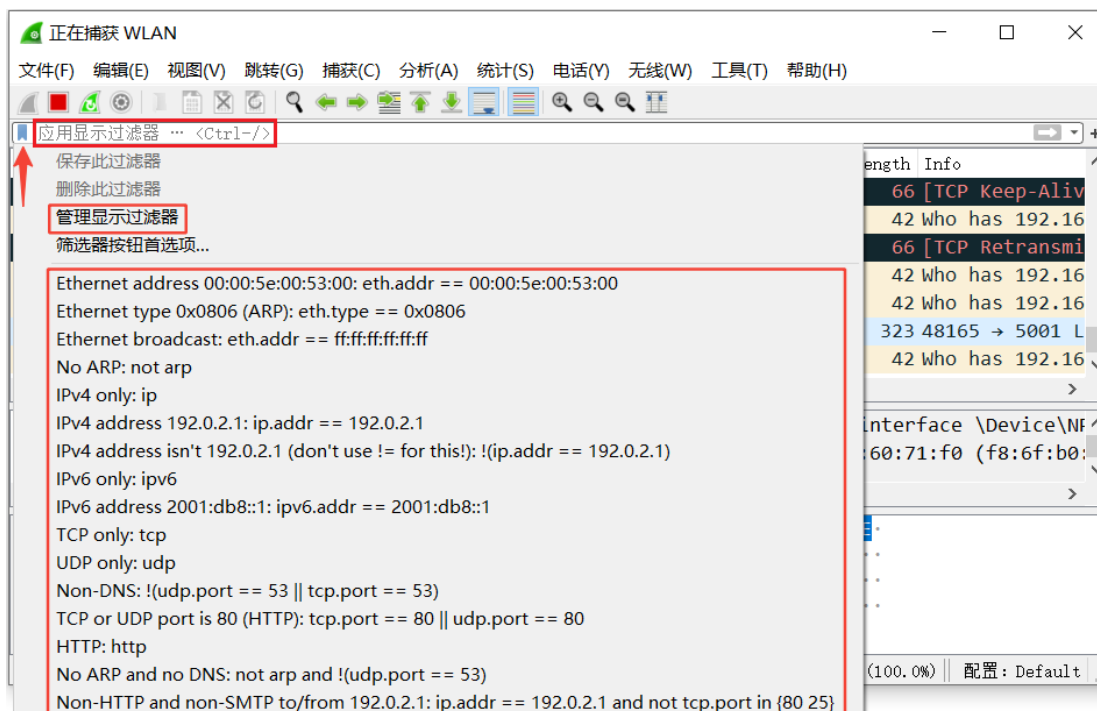
3.3 Configuring a Display Filter

A display filter is used to filter captured data packets.

Step 1 Above the packet list, there is a display filter text box, where you can directly enter a filter expression. Click **Expression** on the right of the text box to open the expression editor. Supported expressions are then displayed in the box, as shown in Figure 13.



Figure 13 Display filter



Step 2 Understand the syntax of display filters. Figure 14 shows the syntax.

Figure 14 Syntax of display filters

Syntax	Protocol	String1	String2	Comparison Operator	Value	Logical Operation	Other Expression
Example	ftp	passive	ip	==	10.1.1.1	xor	icmp.type

- **Protocol:** The supported protocols can be viewed in the editor in Figure 14. Protocols from Layer 2 to Layer 7 of the OSI model are supported.
- **String1** and **String2:** indicate subclasses of a protocol, which are optional.
- **Comparison operators:** Comparison operators and logical operators can be used.

Examples:

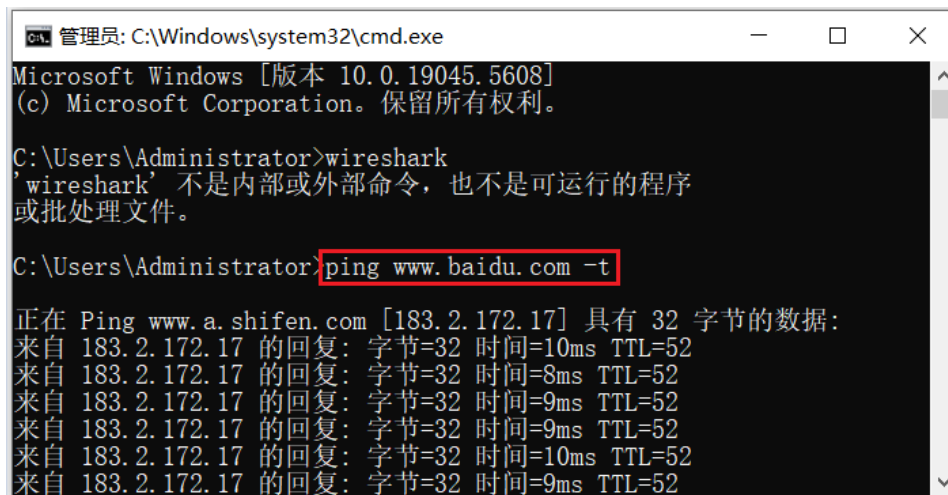
- **tcp.dstport 80 xor tcp.srcport 1025:** displays packets whose destination port is TCP port 80 or packets whose source port is port 1025. It excludes packets that meet both conditions.
- **snmp || dns || icmp:** displays SNMP, DNS, or ICMP packets.
- **ip.addr == 10.1.1.1:** displays packets whose source or destination IP address is 10.1.1.1.
- **ip.src != 10.1.2.3 or ip.dst != 10.4.5.6:** displays packets whose source IP address is not 10.1.2.3 or destination IP address is not 10.4.5.6.
- **ip.src != 10.1.2.3 and ip.dst != 10.4.5.6:** displays packets whose source IP address is not 10.1.2.3 and destination IP address is not 10.4.5.6.
- **tcp.port == 25:** displays packets whose source or destination port is TCP port 25.
- **tcp.dstport == 25:** displays packets whose destination port is TCP port 25.

- **tcp.flags:** displays packets with TCP flags.
- **tcp.flags.syn == 0x02:** displays packets carrying the TCP SYN flag.

If the syntax of the expression entered in the filter expression editor is correct, the background turns green. If the background is red, the expression is incorrect.

Step 3 Configure a display filter to display only ICMP packets. In the cmd window of the PC, ping a specific address, such as **www.baidu.com**, as shown in Figure 15.

Figure 15 Initiating a ping operation



```

C:\Windows\system32\cmd.exe
Microsoft Windows [版本 10.0.19045.5608]
(c) Microsoft Corporation。保留所有权利。

C:\Users\Administrator>wireshark
'wireshark' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

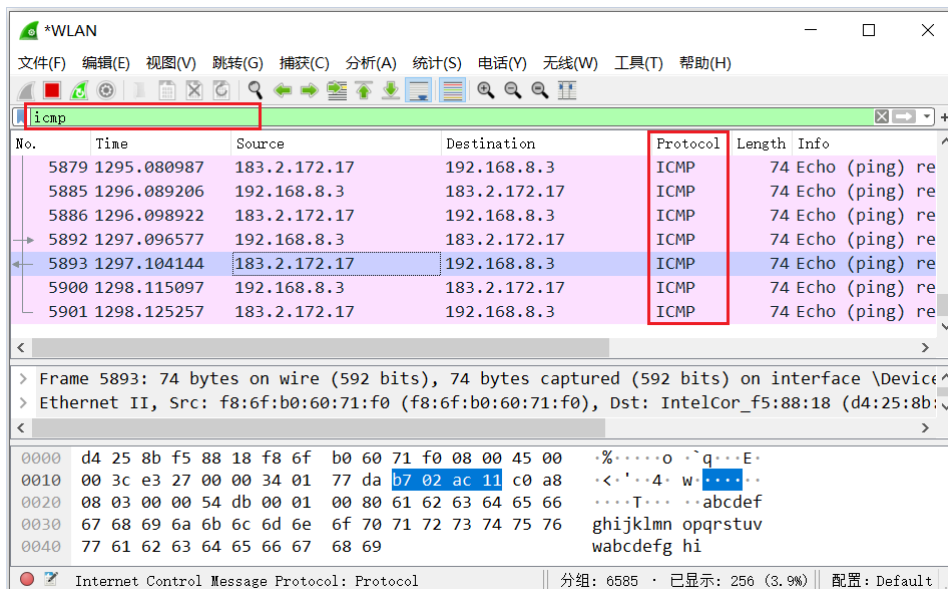
C:\Users\Administrator>ping www.baidu.com -t

正在 Ping www.a.shifen.com [183.2.172.17] 具有 32 字节的数据:
来自 183.2.172.17 的回复: 字节=32 时间=10ms TTL=52
来自 183.2.172.17 的回复: 字节=32 时间=8ms TTL=52
来自 183.2.172.17 的回复: 字节=32 时间=9ms TTL=52
来自 183.2.172.17 的回复: 字节=32 时间=9ms TTL=52
来自 183.2.172.17 的回复: 字节=32 时间=10ms TTL=52
来自 183.2.172.17 的回复: 字节=32 时间=9ms TTL=52

```

Enter **icmp** in the display filter, as shown in Figure 16.

Figure 16 Setting a display filter



No.	Time	Source	Destination	Protocol	Length	Info
5879	1295.080987	183.2.172.17	192.168.8.3	ICMP	74	Echo (ping) re
5885	1296.089206	192.168.8.3	183.2.172.17	ICMP	74	Echo (ping) re
5886	1296.098922	183.2.172.17	192.168.8.3	ICMP	74	Echo (ping) re
5892	1297.096577	192.168.8.3	183.2.172.17	ICMP	74	Echo (ping) re
5893	1297.104144	183.2.172.17	192.168.8.3	ICMP	74	Echo (ping) re
5900	1298.115097	192.168.8.3	183.2.172.17	ICMP	74	Echo (ping) re
5901	1298.125257	183.2.172.17	192.168.8.3	ICMP	74	Echo (ping) re

Frame 5893: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF{...}

Ethernet II, Src: f8:6f:b0:60:71:f0 (f8:6f:b0:60:71:f0), Dst: IntelCor_f5:88:18 (d4:25:8b:...

```

0000  d4 25 8b f5 88 18 f8 6f  b0 60 71 f0 08 00 45 00  .%. . . . . o .`q...E.
0010  00 3c e3 27 00 00 34 01  77 da b7 02 ac 11 c0 a8  <...4. w. ....
0020  08 03 00 00 54 db 00 01  00 80 61 62 63 64 65 66  . . . . T . . . abcdef
0030  67 68 69 6a 6b 6c 6d 6e  6f 70 71 72 73 74 75 76  ghijklmn opqrstuv
0040  77 61 62 63 64 65 66 67  68 69                wabcdefg hi

```

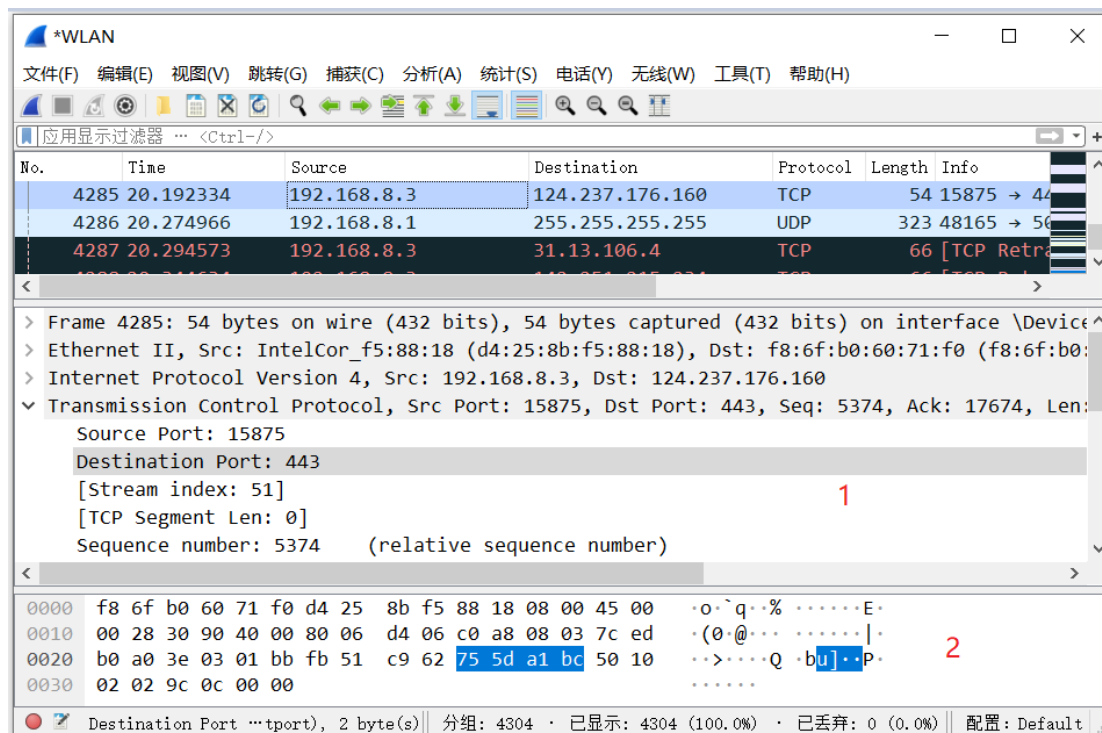
Internet Control Message Protocol: Protocol | 分组: 6585 · 已显示: 256 (3.9%) | 配置: Default



3.4 Data Packet Analysis

Step 1 Select a line in the packet list. The details of the corresponding packet are displayed in two areas, as shown in Figure 17.

Figure 17 Details of a data packet



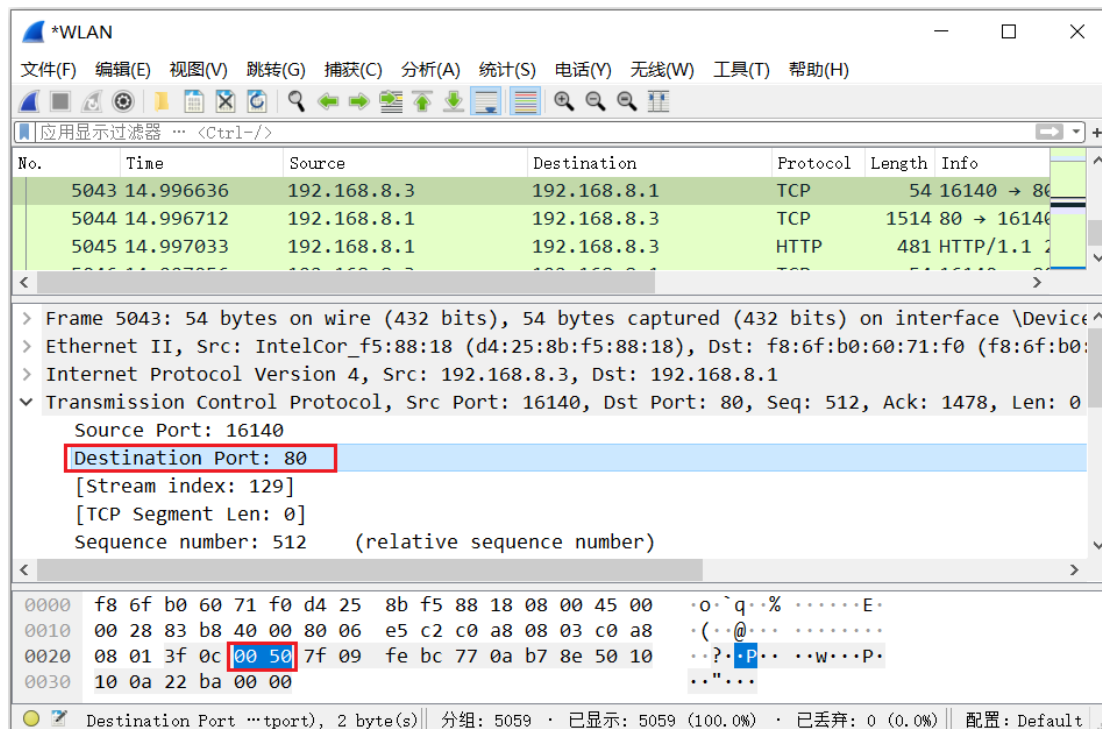
In Figure 17, area 1 displays detailed protocol breakdown of the selected packet, and area 2 displays the hexadecimal representation of the selected packet. Area 1 includes the following fields:

- **Frame:** overview of the data frame at the physical layer
- **Ethernet II:** Ethernet frame header information at the data link layer
- **Internet Protocol Version 4:** IP packet header information at the network layer
- **Transmission Control Protocol:** header information of the data segment at the transport layer. In this example, it is TCP.
- **Hypertext Transfer Protocol:** application layer information. In this example, it is HTTPS.

When you click a field in area 1, the corresponding data item is displayed in area 2, as shown in Figure 18.

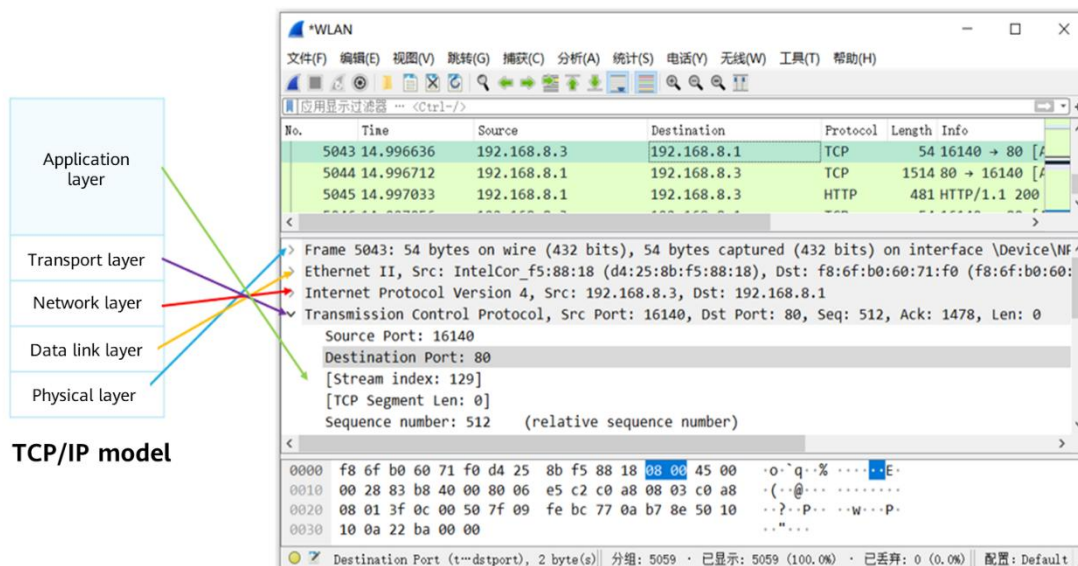


Figure 18 Data item



Step 2 Check the mappings between layers in the TCP/IP model and data packet analysis areas in Wireshark, as shown in Figure 19.

Figure 19 Mappings between layers in the TCP/IP model and data packet analysis areas in Wireshark



Step 3 Check the Ethernet_II frame header format at the data link layer. Figure 20 shows the frame format.



Figure 20 Ethernet_II frame header format

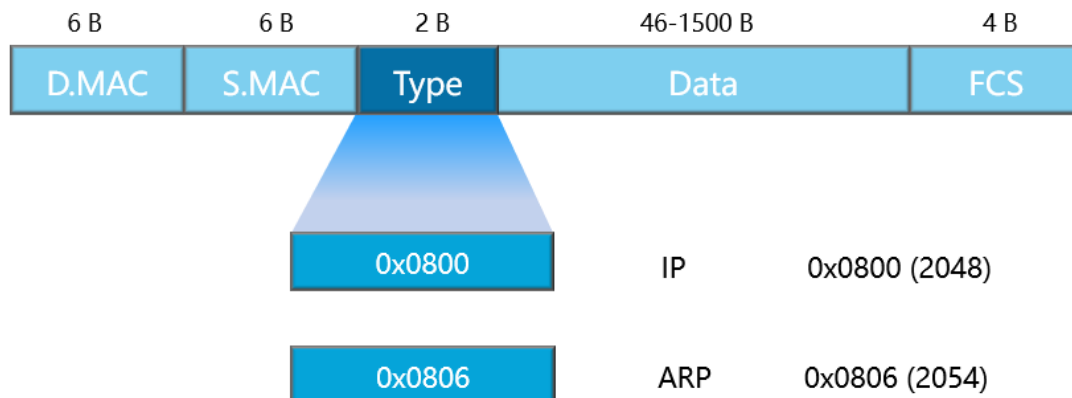
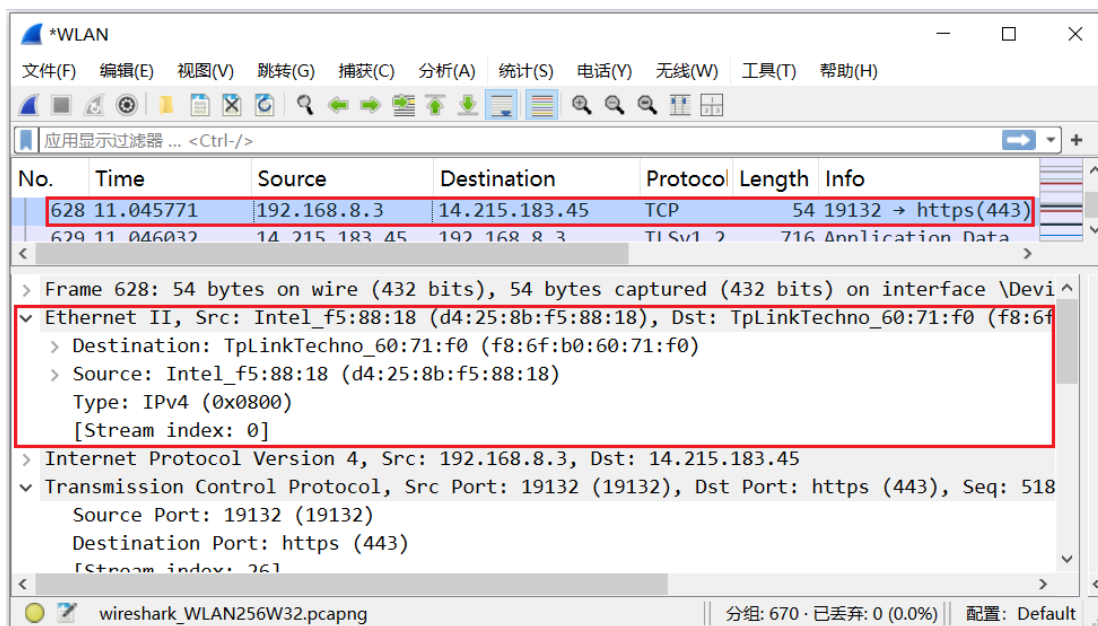


Figure 21 shows the frame header displayed in Wireshark.

Figure 21 Frame header in Wireshark



Step 4 Check the IPv4 header format at the network layer. Figure 22 shows the header format.



Figure 22 IPv4 header format

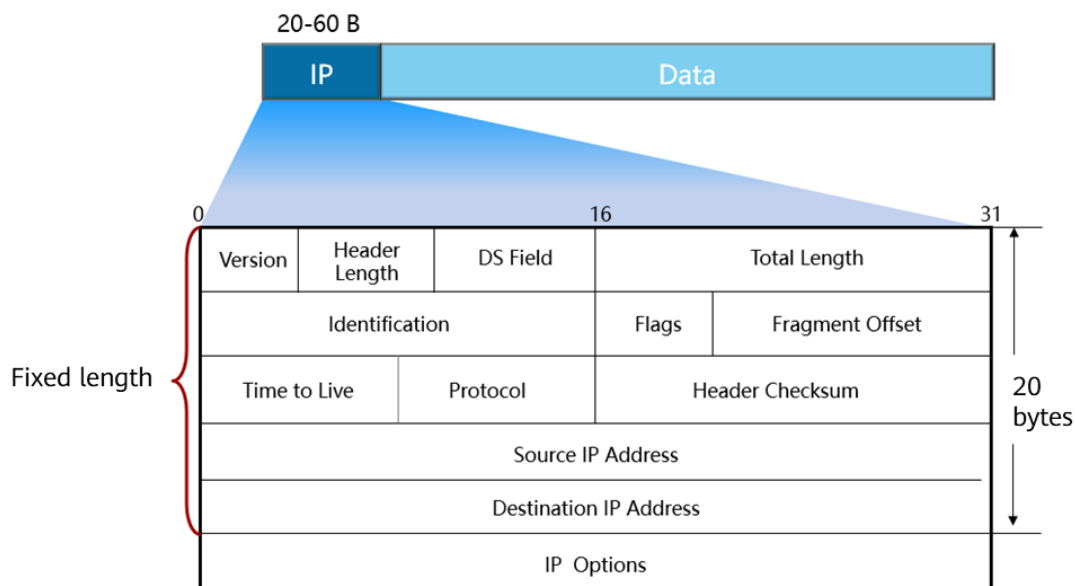
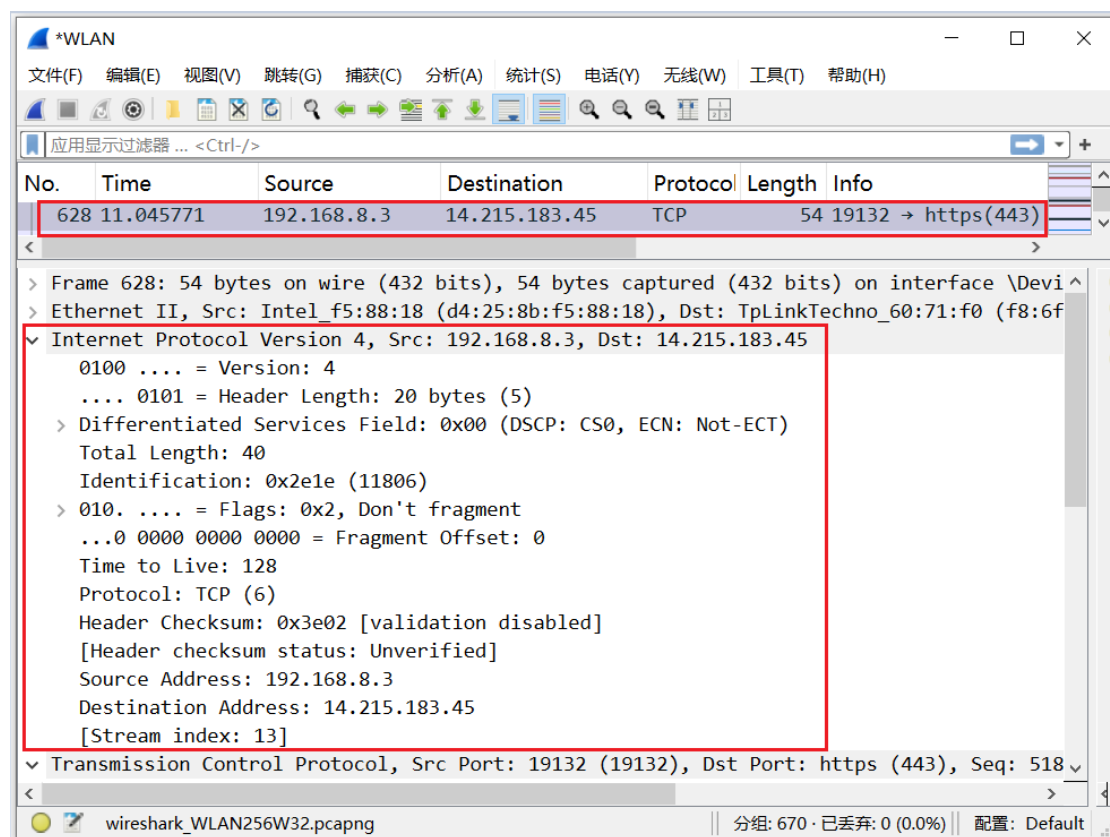


Figure 23 shows the IPv4 header displayed in Wireshark.

Figure 23 IPv4 header in Wireshark





Step 5 Check the TCP header format. Figure 24 shows the header format.

Figure 24 TCP header format

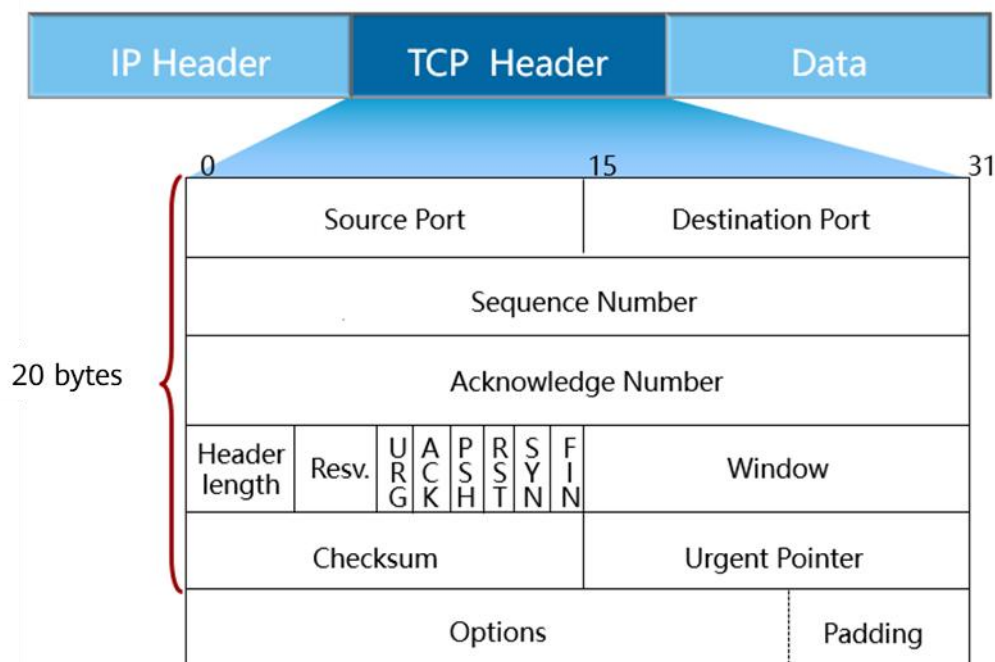
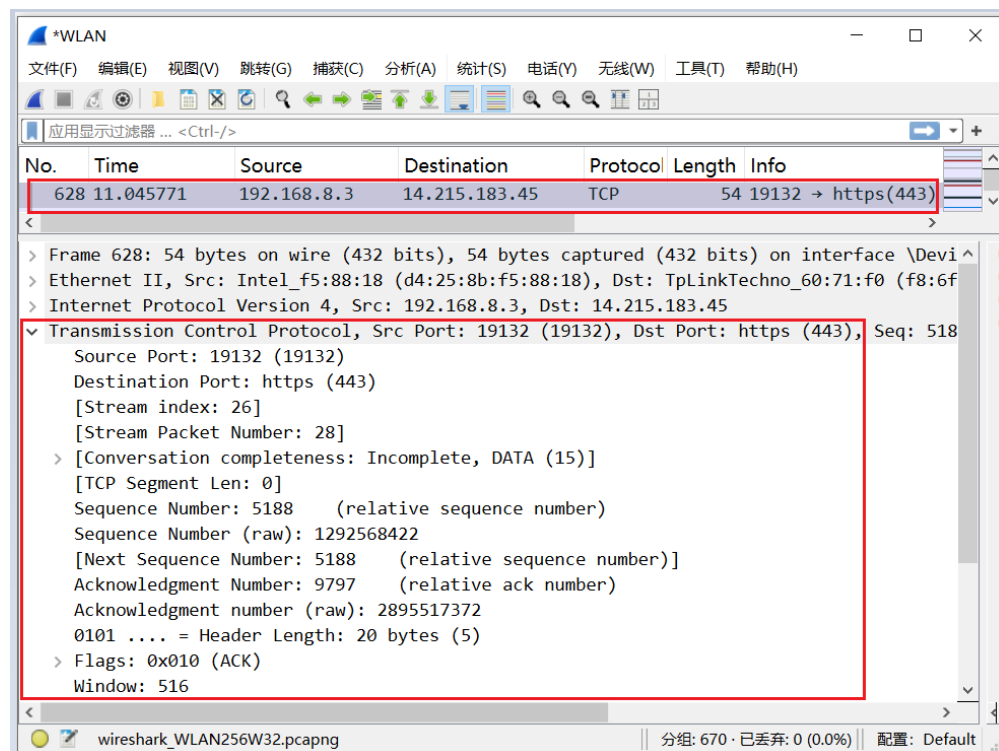


Figure 25 shows the TCP header displayed in Wireshark.

Figure 25 TCP header in Wireshark





Step 6 Check the UDP header format. Figure 26 shows the header format.

Figure 26 UDP header format

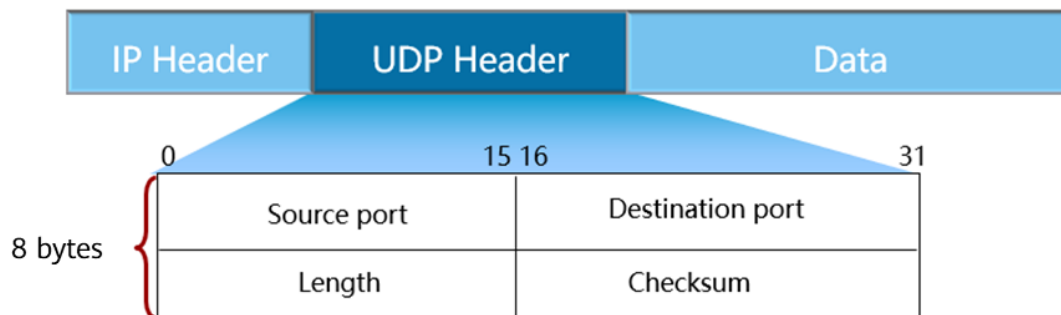
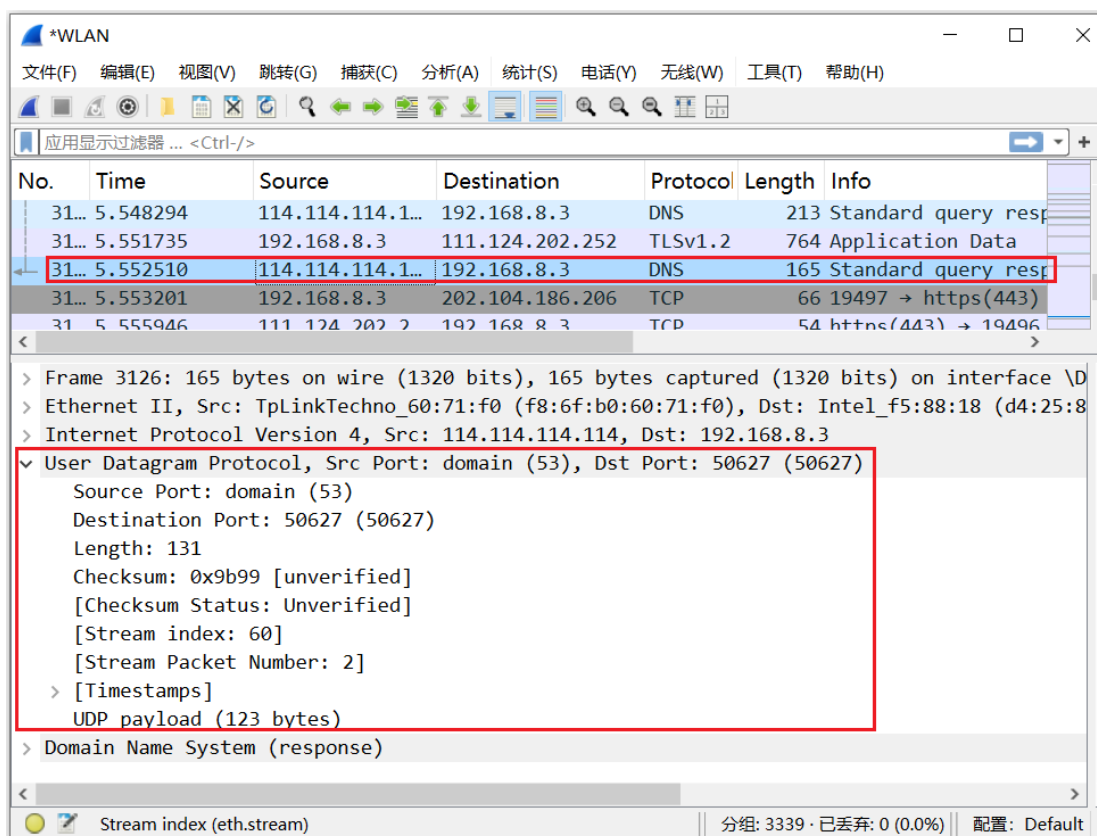


Figure 27 shows the UDP header displayed in Wireshark.

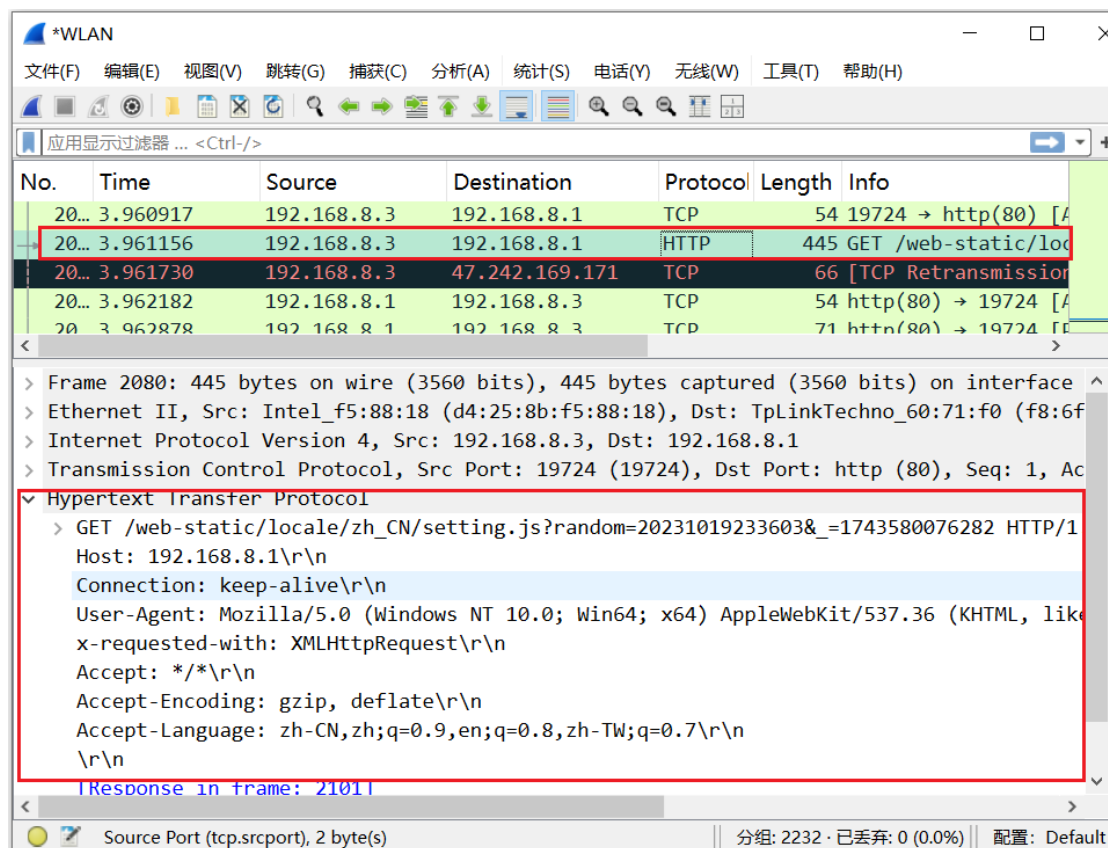
Figure 27 UDP header in Wireshark



Step 7 Check the HTTP header displayed in Wireshark, as shown in Figure 28.



Figure 28 HTTP header in Wireshark

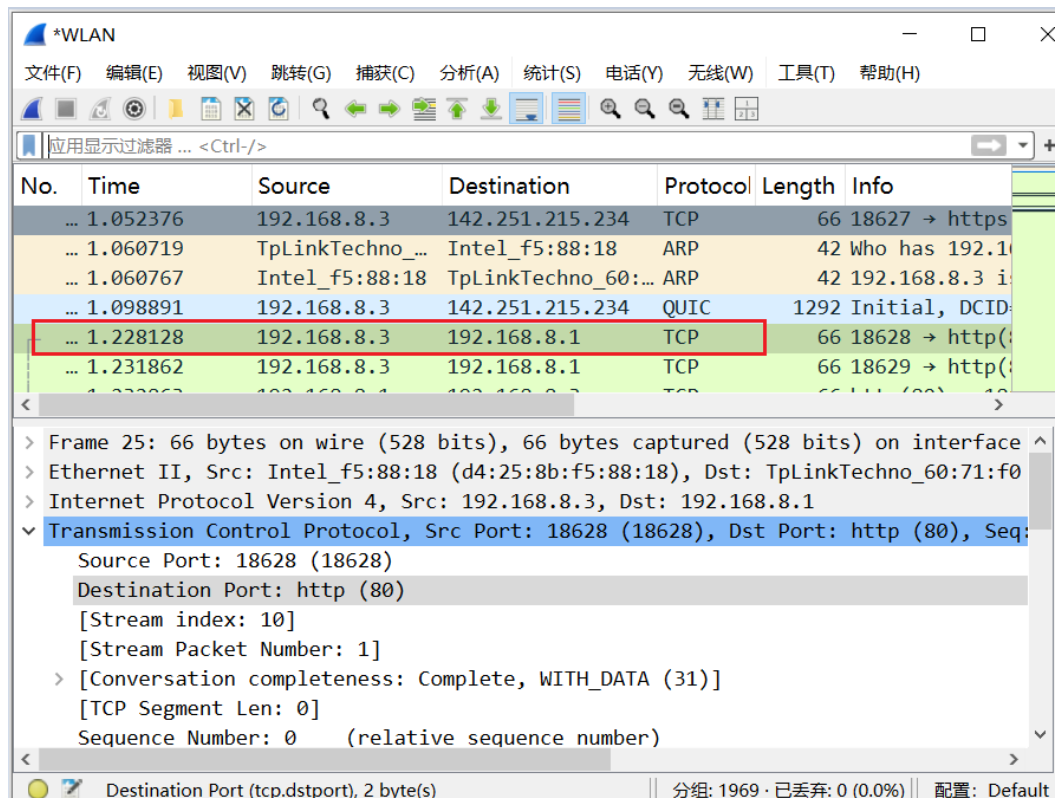


3.5 TCP Three-Way Handshake

- Step 1** Open Wireshark to capture packets and enter 192.168.8.1 in the address box of the browser. After packet capture is stopped, the following page is displayed, as shown in Figure 29.

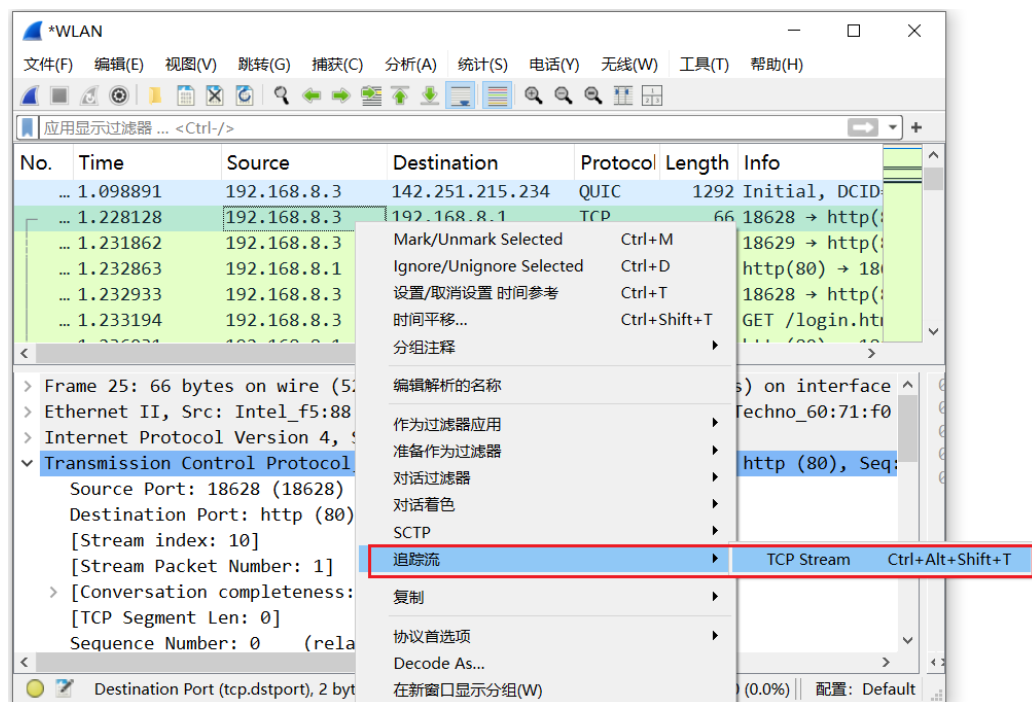


Figure 29 Wireshark packet capture



Step 2 Select a packet, right-click it, and choose **Follow > TCP Stream** from the shortcut menu, as shown in Figure 30.

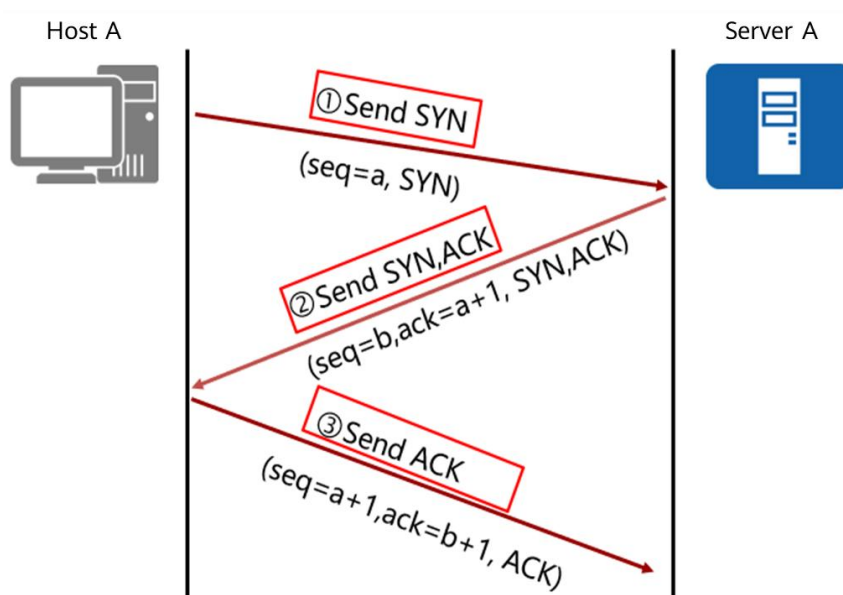
Figure 30 Following a TCP stream





Wireshark then automatically applies a display filter based on the tcp.stream field to display packets exchanged in a TCP three-way handshake, including the HTTP request. Figure 31 shows the process of a TCP three-way handshake.

Figure 31 TCP three-way handshake



Wireshark shows the packets exchanged in a TCP three-way handshake, as shown in Figure 32.

Figure 32 Packets exchanged in a TCP three-way handshake

The screenshot shows a Wireshark packet capture window with the display filter `tcp.stream eq 10`. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
25	1.228128	192.168.8.3	192.168.8.1	TCP	66	18628 → http(80) [SYN] Seq=0
27	1.232863	192.168.8.1	192.168.8.3	TCP	66	http(80) → 18628 [SYN, ACK] Seq=1
28	1.232933	192.168.8.3	192.168.8.1	TCP	54	18628 → http(80) [ACK] Seq=1
29	1.233194	192.168.8.3	192.168.8.1	HTTP	510	GET /login.htm HTTP/1.1
32	1.239972	192.168.8.1	192.168.8.3	TCP	54	http(80) → 18628 [ACK] Seq=1
59	1.886608	192.168.8.1	192.168.8.3	TCP	63	http(80) → 18628 [PSH, ACK] Seq=10
60	1.888089	192.168.8.1	192.168.8.3	TCP	1514	http(80) → 18628 [ACK] Seq=14
61	1.888089	192.168.8.1	192.168.8.3	TCP	1514	http(80) → 18628 [ACK] Seq=29
62	1.888089	192.168.8.1	192.168.8.3	TCP	1514	http(80) → 18628 [ACK] Seq=45
63	1.888210	192.168.8.3	192.168.8.1	TCP	54	18628 → http(80) [ACK] Seq=45
64	1.890395	192.168.8.1	192.168.8.3	TCP	1150	http(80) → 18628 [PSH, ACK] Seq=45
65	1.935848	192.168.8.3	192.168.8.1	TCP	54	18628 → http(80) [ACK] Seq=45

4 Verification

None.



5 Quiz

1. Use a capture filter to capture all packets except ICMP packets.
2. Use a display filter to display packets whose destination port is TCP port 25.
3. What is the relationship between MAC frame and IP datagram lengths? Verify the relationship using your data records.
A MAC frame adds 18 bytes to an IP datagram. An IP datagram is the payload that a MAC frame carries.
4. Will the packet capture workload of Wireshark be reduced after display filters are used?
No. Display filters only affect what is shown in the Wireshark interface, but does not affect packet capture.